

$\mathfrak{su}(3)$ from a Trapped Tetrahedral Defect in the FCC Lattice

Raghu Kulkarni*

SSMTheory Group, IDrive Inc., Calabasas, CA 91302, USA

May 2026

Abstract

The matter paper of the Selection-Stitch Model [1] traces the three QCD color charges to the three skew-edge pairs of the tetrahedron that bounds a trapped extra node in the FCC vacuum, but stops short of the $\mathfrak{su}(3)$ Lie algebra. We close that gap. The skew pairs span a 3-dimensional color space; the quotient $S_4/V_4 \cong S_3$ generated by anchor selection on the defect acts as the Weyl group of $\mathfrak{su}(3)$; and the $U(1)$ phase carried by each Bell-pair valence bond, modulo the diagonal $U(1)$ that gauges to electromagnetism, supplies a rank-2 maximal torus. Two diagonal generators built from the bond phases together with six off-diagonal generators built from the oriented color transitions close on $\mathfrak{su}(3)$ with the Cartan-Weyl structure constants of the A_2 root system, verified by explicit computation of all 28 commutators. The fundamental representation $\mathbf{3}$, the rank, the Weyl group, the maximal torus, and the full root system emerge from the defect geometry without adjustable inputs. The QCD coupling g_s , its running, Λ_{QCD} , and other dynamical features lie outside the static algebraic content of the construction.

1 Introduction

The matter paper of the SSM derives the count of QCD colors from a single combinatorial fact: the six edges of the tetrahedron K_4 partition uniquely into three pairs of vertex-disjoint (skew) edges, with no fourth such partition. The matter paper assigns one color to each skew pair and identifies the count, $C(4,2)/2 = 3$, with the dimension of the fundamental representation of QCD's gauge group. It is also explicit that this is a derivation of the representation, not of the Lie algebra:

“We do not claim to derive the $SU(3)$ Lie algebra from the lattice geometry; we claim that the representation of color charges on a baryon-like defect is uniquely fixed at three by the combinatorial identity $C(4,2)/2 = 3$.” [1, Section 5.2]

The present paper supplies the missing step. The defect carries enough structure to reconstruct the eight-dimensional algebra: three pieces, each grounded in a different feature of the geometry, combine to give exactly $\mathfrak{su}(3)$. The first piece is combinatorial (the skew-pair partition of K_4). The second piece is symmetry-theoretic (the quotient S_4/V_4 from anchor selection). The third piece is continuous: a $U(1)$ phase per valence bond, with the diagonal phase gauged to electromagnetism. The rank, the maximal torus, the Weyl group, the root system, and the structure constants all fall out together.

The construction is static and algebraic. The dynamical content of QCD — coupling strength, running, asymptotic freedom, Λ_{QCD} — belongs to the bond energetics of the lattice, which we do not address.

*raghu@idrive.com

2 The Defect

We work with the trapped tetrahedral defect of the matter paper [1, Section 3.2]. An extra node sits at the centroid of a tetrahedral void of the FCC lattice, bonded to four vertices A, B, C, D that form a regular tetrahedron of edge length L . The four bonds split into one anchor and three valence bonds when one of the four directions $\hat{r}_A, \hat{r}_B, \hat{r}_C, \hat{r}_D$ is selected as the bulk-coupling channel; this is the matter paper’s anchor-selection step that breaks $S_4 \rightarrow S_3$ on the bond directions.

3 Three colors as skew pairs

3.1 The skew-pair partition

The complete graph K_4 on $\{A, B, C, D\}$ has $\binom{4}{2} = 6$ edges. A pair of edges is *skew* if the edges share no vertex; for K_4 this means the four endpoints are all distinct. Any skew pair specifies a partition of $\{A, B, C, D\}$ into two disjoint unordered pairs (the endpoints of each edge). Conversely, any partition of four elements into two unordered pairs determines a skew pair of edges. The number of such partitions is

$$\frac{1}{2} \binom{4}{2} = 3, \quad (1)$$

where the factor of $1/2$ accounts for the symmetry between the two pairs. The three skew pairs of K_4 are

$$P_1 = \{AB, CD\}, \quad P_2 = \{AC, BD\}, \quad P_3 = \{AD, BC\}. \quad (2)$$

These three pairs partition the six edges of K_4 into three groups of two, and there is no fourth such partition — $\binom{4}{2}/2 = 3$ is exact, not an approximation. Figure 1 displays them. Following the matter paper [1], we identify the three skew pairs with the three QCD color charges.

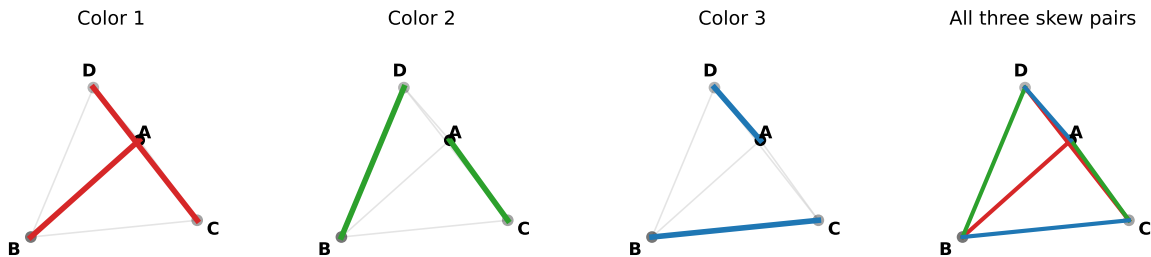


Figure 1: The three skew-edge pairs of the bounding tetrahedron K_4 . Each pair consists of two edges that share no vertex; together the three pairs partition the six edges, and there is no fourth such partition. The first three panels show one pair at a time (red, green, blue); the fourth panel overlays all three. The matter paper [1] identifies each pair with one QCD color charge.

3.2 The color Hilbert space

We package the three colors into a Hilbert space

$$\mathcal{H}_C \equiv \mathbb{C}^3 = \text{span}_{\mathbb{C}}(|1\rangle, |2\rangle, |3\rangle), \quad |i\rangle \leftrightarrow P_i, \quad (3)$$

on which we will construct the $\mathfrak{su}(3)$ generators. The basis vectors $|1\rangle, |2\rangle, |3\rangle$ are the three skew-pair states of the trapped node; the inner product is the standard Hermitian inner product on $\mathbb{C}^3$. The states are by construction orthonormal: $\langle i|j\rangle = \delta_{ij}$.

4 The Weyl group from anchor selection

4.1 The action of S_4 on the skew pairs

The full permutation group of $\{A, B, C, D\}$ is the symmetric group S_4 of order $4! = 24$. Each element $\sigma \in S_4$ acts on edges by permuting endpoints, $\sigma(\{X, Y\}) = \{\sigma(X), \sigma(Y)\}$, and on skew pairs by acting on each edge:

$$\sigma(P) = \sigma(\{e_1, e_2\}) = \{\sigma(e_1), \sigma(e_2)\}. \quad (4)$$

Since the property of being a skew pair is preserved under permutations (vertex-disjointness is permutation-invariant), $\sigma(P_i)$ is again a skew pair. We obtain a homomorphism

$$\pi : S_4 \rightarrow \text{Sym}(\{P_1, P_2, P_3\}) \cong S_3, \quad \sigma \mapsto (P_i \mapsto \sigma(P_i)). \quad (5)$$

Proposition 1. *The kernel of π is the Klein four-group*

$$V_4 = \{e, (AB)(CD), (AC)(BD), (AD)(BC)\} \subset S_4. \quad (6)$$

Proof. We compute $\pi(g)$ for each $g \in S_4$ by direct enumeration, focusing on the four elements of (6).

- $\pi(e)$ fixes each pair: $e(P_i) = P_i$. Trivially in the kernel.
- $\pi((AB)(CD))$: applying $(AB)(CD)$ to $P_1 = \{AB, CD\}$ gives $\{(AB)(CD)(\{A, B\}), (AB)(CD)(\{C, D\})\} = \{\{B, A\}, \{D, C\}\} = \{AB, CD\} = P_1$. Applying it to $P_2 = \{AC, BD\}$ gives $\{\{B, D\}, \{A, C\}\} = P_2$. Similarly $P_3 \mapsto P_3$. In the kernel.
- $\pi((AC)(BD))$ and $\pi((AD)(BC))$: by the same computation, each fixes P_1, P_2, P_3 as sets. In the kernel.

For any other element of S_4 , $\pi(g) \neq e$. For example, $\pi((AB))$: applying (AB) to $P_2 = \{AC, BD\}$ gives $\{BC, AD\} = P_3$, so (AB) sends P_2 to P_3 , and is not in the kernel. The four elements of V_4 are therefore the complete kernel. $\square$

By the first isomorphism theorem,

$$S_4/V_4 \cong \text{Im}(\pi) = S_3. \quad (7)$$

The image is all of S_3 because every transposition of skew pairs $(P_i P_j)$ is realized by some element of S_4 (e.g., the swap $(P_1 P_2)$ is induced by (AC)).

Anchor selection in the matter paper picks one of the four vertices — without loss of generality A — as the channel coupling the defect to the bulk. The stabilizer of A in S_4 is the symmetric group on $\{B, C, D\}$, isomorphic to S_3 . Composing the inclusion $S_3 \hookrightarrow S_4$ with π gives an isomorphism

$$S_3 \xrightarrow{\sim} S_4/V_4, \quad (8)$$

since no non-trivial permutation of $\{B, C, D\}$ lies in V_4 . The residual symmetry of the defect after anchor selection is therefore precisely the quotient $S_4/V_4 \cong S_3$.

4.2 Identification with the Weyl group of $\mathfrak{su}(3)$

The Weyl group of a Lie algebra $\mathfrak{g}$ is the group generated by reflections through the hyperplanes orthogonal to the roots; it is the symmetry group of the root system. For $\mathfrak{su}(3)$ the root system is A_2 , and reflections through any of the three pairs of opposite roots generate

$$W(A_2) = S_3, \quad (9)$$

the symmetric group on three letters [3, Section 10.3]. The defect's residual S_3 symmetry is therefore identical to the Weyl group of $\mathfrak{su}(3)$ — the same group acting on the same three objects (the three colors / weight states), not an analog or a finite approximation.

5 The maximal torus from bond phases

5.1 The $U(1)$ phase of a Bell pair

A Bell-pair entanglement state on two qubits has the general form

$$|\Phi(\phi)\rangle = \frac{1}{\sqrt{2}}(|00\rangle + e^{i\phi}|11\rangle), \quad (10)$$

parametrized by a single real phase $\phi \in [0, 2\pi)$ that distinguishes inequivalent Bell pairs up to local unitary equivalence [5]. The matter paper's identification of FCC bonds with Bell-pair entanglement [1, Section 2.3] therefore assigns one continuous phase $\phi \in U(1)$ to each bond.

For the trapped tetrahedral defect, after anchor selection the three valence bonds $\hat{r}_B, \hat{r}_C, \hat{r}_D$ each carry an independent phase $\phi_B, \phi_C, \phi_D \in U(1)$. The full phase space is the 3-torus

$$T^3 = U(1)_B \times U(1)_C \times U(1)_D. \quad (11)$$

5.2 Quotienting the diagonal $U(1)$

The diagonal subgroup

$$U(1)_{\text{diag}} = \{(e^{i\theta}, e^{i\theta}, e^{i\theta}) : \theta \in [0, 2\pi)\} \subset T^3 \quad (12)$$

shifts all three valence-bond phases by the same amount. Its action on the trapped node, viewed as a localized state in $\mathcal{H}_C$, is by an overall phase $|\psi\rangle \mapsto e^{i\theta}|\psi\rangle$, since each color basis state $|i\rangle$ picks up the same factor. This overall phase is unobservable on a closed system, but it couples to charged probes in the standard way of a $U(1)$ gauge symmetry. The matter paper [1, Section 4] identifies this diagonal $U(1)$ with electromagnetism, assigning it the role of fixing the trapped node's electric charge (the integer winding $W = w_B + w_C + w_D$ of the matter paper's Eq. 14 is the conserved charge associated with $U(1)_{\text{diag}}$).

Quotienting $U(1)_{\text{diag}}$ out of T^3 leaves

$$T^2 \equiv T^3 / U(1)_{\text{diag}} = \{(\phi_B, \phi_C, \phi_D) : \phi_B + \phi_C + \phi_D \equiv 0 \pmod{2\pi}\}. \quad (13)$$

This is a two-dimensional torus, since the constraint $\phi_B + \phi_C + \phi_D = 0$ removes one direction from the original T^3 . The dimension matches the rank of $\mathfrak{su}(3)$.

Lemma 1. *The rank of $\mathfrak{su}(3)$, defined as the dimension of any maximal abelian subalgebra of diagonalizable elements, equals 2.*

Proof. The diagonal traceless 3×3 Hermitian matrices form a 2-dimensional abelian subalgebra of $\mathfrak{su}(3)$: any two diagonal matrices commute, and the traceless condition imposes one constraint on the three diagonal entries, leaving 2 free parameters. This subalgebra is maximal: any traceless Hermitian matrix that commutes with all diagonal matrices must itself be diagonal, since the eigenspaces of a generic diagonal matrix are 1-dimensional. The dimension of this maximal abelian subalgebra is the rank. $\square$

5.3 Cartan generators

A natural basis of infinitesimal generators of T^2 , expressed as diagonal traceless Hermitian matrices on $\mathcal{H}_C$, is

$$H_1 = \text{diag}(1, -1, 0), \quad (14)$$

$$H_2 = \frac{1}{\sqrt{3}} \text{diag}(1, 1, -2). \quad (15)$$

Verification of the basis properties.

1. *Tracelessness.* $\text{tr}(H_1) = 1 + (-1) + 0 = 0$. $\text{tr}(H_2) = (1 + 1 - 2)/\sqrt{3} = 0$. Both are traceless; equivalently, both lie in $\mathfrak{su}(3)$ rather than $\mathfrak{u}(3)$, having quotiented the diagonal $U(1)$.
2. *Commutativity.* Diagonal matrices commute: $[H_1, H_2] = H_1 H_2 - H_2 H_1 = 0$, as can be verified entry-wise since $(H_1 H_2)_{ii} = (H_1)_{ii} (H_2)_{ii} = (H_2 H_1)_{ii}$.
3. *Linear independence.* The vectors $(1, -1, 0)$ and $\frac{1}{\sqrt{3}}(1, 1, -2)$ are linearly independent in $\mathbb{R}^3$ (their cross product is non-zero), so H_1, H_2 span the 2-dimensional traceless diagonal subalgebra.

Geometric meaning. The infinitesimal action of H_1 on a color state $|j\rangle$ is $H_1|j\rangle = (H_1)_{jj}|j\rangle$, so $|1\rangle$ acquires phase $+1$, $|2\rangle$ phase -1 , and $|3\rangle$ phase 0 . This is the difference of valence-bond phases on bonds carrying colors 1 and 2 (with color 3 unaffected). Similarly, H_2 assigns phase $1/\sqrt{3}$ to colors 1 and 2 and phase $-2/\sqrt{3}$ to color 3, measuring the imbalance between $\frac{1}{2}(\phi_1 + \phi_2)$ and ϕ_3 .

Why $1/\sqrt{3}$. The factor $1/\sqrt{3}$ in H_2 is the unique normalization that makes H_1, H_2 orthonormal under the Killing form $K(X, Y) = 2 \text{tr}(XY)$ on $\mathfrak{su}(3)$. The trace evaluates as

$$\text{tr}(H_1^2) = 1 + 1 + 0 = 2, \quad (16)$$

$$\text{tr}(H_2^2) = \frac{1}{3}(1 + 1 + 4) = 2, \quad (17)$$

$$\text{tr}(H_1 H_2) = \frac{1}{\sqrt{3}}(1 \cdot 1 + (-1) \cdot 1 + 0 \cdot (-2)) = 0, \quad (18)$$

giving $K(H_i, H_j) = 4\delta_{ij}$ (a non-zero scalar multiple of the identity). Without the $1/\sqrt{3}$ factor the algebra is unchanged, but the roots in the (H_1, H_2) plane have unequal lengths and the A_2 angles are obscured.

6 Root generators from color transitions

6.1 Definition

A root vector of $\mathfrak{su}(3)$ raises or lowers between weight spaces (eigenspaces of the Cartan subalgebra). In the color basis, the off-diagonal matrix units

$$E_{ij} = |i\rangle\langle j|, \quad i \neq j, \quad i, j \in \{1, 2, 3\}, \quad (19)$$

take the trapped node from color j to color i . As 3×3 matrices,

$$E_{12} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad E_{13} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad E_{23} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad (20)$$

and their adjoints $F_{ij} \equiv E_{ij}^\dagger = E_{ji}$,

$$F_{12} = E_{12}^\dagger = E_{21} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (21)$$

and similarly for F_{13}, F_{23} . The number of distinct ordered pairs (i, j) with $i \neq j$ in $\{1, 2, 3\}$ is $3 \times 2 = 6$, giving six off-diagonal generators

$$\{E_{12}, E_{13}, E_{23}, F_{12}, F_{13}, F_{23}\}. \quad (22)$$

6.2 Counting and dimension

Adding the two Cartan generators H_1, H_2 to the six off-diagonal E_{ij}, F_{ij} yields eight operators. The dimension of $\mathfrak{su}(3)$ is

$$\dim_{\mathbb{R}} \mathfrak{su}(3) = n^2 - 1 = 9 - 1 = 8 \quad (n = 3), \quad (23)$$

where the -1 subtracts the trace, and the count matches.

Vertex versus propagator. An operator that switches a quark's color is, in QCD, a gluon emission/absorption vertex. The E_{ij} and F_{ij} defined here are the algebraic content of that vertex: the local action of a gluon on the trapped node, expressed as a transition between skew-pair labels. They are not the propagating gluon itself. A propagating gluon is a degree of freedom living on the FCC bulk between defects, with a dispersion relation, a mass shell, and dynamics set by the bond energetics. The present paper constructs only the vertex algebra; the propagator and its dynamics are the natural follow-on question we defer in Section 9.

7 Closure on $\mathfrak{su}(3)$

We verify by explicit matrix computation that the eight operators close on a Lie algebra, and that the algebra is $\mathfrak{su}(3)$. The full table of all 28 commutators is in Appendix A; here we show the calculations needed to identify the A_2 root system.

7.1 Cartan action on root vectors

For each root vector E_{ij} , the commutator with a Cartan generator H has the form

$$[H, E_{ij}] = \alpha_{ij}(H) E_{ij}, \quad (24)$$

with the eigenvalue $\alpha_{ij}(H)$ depending linearly on H . We compute $[H_1, E_{12}]$ in detail as a worked example; the rest follow by symmetric calculation. With $H_1 = \text{diag}(1, -1, 0)$ and E_{12} from Eq. (20),

$$H_1 E_{12} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (25)$$

$$E_{12}H_1 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (26)$$

$$[H_1, E_{12}] = H_1 E_{12} - E_{12} H_1 = \begin{pmatrix} 0 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = 2E_{12}. \quad (27)$$

Hence the eigenvalue is $\alpha_{12}(H_1) = 2$. The same calculation method applied to all combinations of Cartan with root vector gives the eigenvalue table

	E_{12}	E_{13}	E_{23}	F_{12}	F_{13}	F_{23}
H_1	+2	+1	-1	-2	-1	+1
H_2	0	+ $\sqrt{3}$	+ $\sqrt{3}$	0	- $\sqrt{3}$	- $\sqrt{3}$

(28)

The columns of (28) are the root coordinates $(\alpha(H_1), \alpha(H_2))$ in the (H_1, H_2) plane:

$$\alpha_{12} = (2, 0), \quad \alpha_{13} = (1, \sqrt{3}), \quad \alpha_{23} = (-1, \sqrt{3}), \quad (29)$$

together with $-\alpha_{12}, -\alpha_{13}, -\alpha_{23}$, plotted in Figure 2.

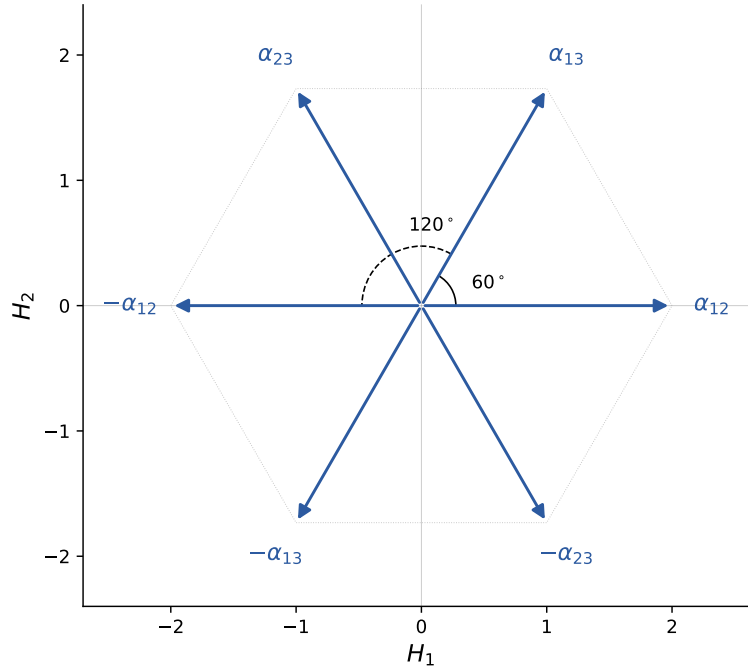


Figure 2: The A_2 root system in the (H_1, H_2) plane. The six roots, recovered from the eigenvalues of H_1, H_2 on the color transitions E_{ij} , sit on a regular hexagon with $|\alpha|^2 = 4$ and 60° between adjacent roots. This is the root system of $\mathfrak{su}(3)$.

7.2 Root lengths and angles

Each root has length squared

$$|\alpha_{12}|^2 = 2^2 + 0^2 = 4, \quad (30)$$

$$|\alpha_{13}|^2 = 1^2 + (\sqrt{3})^2 = 1 + 3 = 4, \quad (31)$$

$$|\alpha_{23}|^2 = (-1)^2 + (\sqrt{3})^2 = 1 + 3 = 4. \quad (32)$$

All six roots therefore have $|\alpha|^2 = 4$ in this normalization. The inner products between distinct positive roots are

$$\alpha_{12} \cdot \alpha_{13} = 2 \cdot 1 + 0 \cdot \sqrt{3} = 2, \quad (33)$$

$$\alpha_{12} \cdot \alpha_{23} = 2 \cdot (-1) + 0 \cdot \sqrt{3} = -2, \quad (34)$$

$$\alpha_{13} \cdot \alpha_{23} = 1 \cdot (-1) + \sqrt{3} \cdot \sqrt{3} = -1 + 3 = 2, \quad (35)$$

giving cosines

$$\cos \theta_{12,13} = \frac{2}{\sqrt{4}\sqrt{4}} = \frac{1}{2}, \quad \cos \theta_{12,23} = -\frac{1}{2}, \quad \cos \theta_{13,23} = \frac{1}{2}, \quad (36)$$

hence angles $60^\circ, 120^\circ, 60^\circ$. This is the A_2 root system: the unique rank-2 simply-laced root system with all roots of equal length and angles in $\{60^\circ, 120^\circ, 180^\circ\}$. By the classification of complex semisimple Lie algebras [3, 4, Section 11], the algebra with root system A_2 is $\mathfrak{su}(3)$ uniquely (up to isomorphism over $\mathbb{C}$, with real form $\mathfrak{su}(3)$ fixed by Hermiticity of the generators).

7.3 Root-Cartan generation

The commutators of root vectors with their negatives recover the Cartan subalgebra. We compute $[E_{12}, F_{12}]$ explicitly:

$$E_{12}F_{12} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (37)$$

$$F_{12}E_{12} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (38)$$

$$[E_{12}, F_{12}] = E_{12}F_{12} - F_{12}E_{12} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} = H_1. \quad (39)$$

The analogous calculations for the other two pairs give

$$[E_{13}, F_{13}] = \text{diag}(1, 0, -1) = \frac{1}{2}H_1 + \frac{\sqrt{3}}{2}H_2, \quad (40)$$

$$[E_{23}, F_{23}] = \text{diag}(0, 1, -1) = -\frac{1}{2}H_1 + \frac{\sqrt{3}}{2}H_2, \quad (41)$$

where the right-hand sides are the explicit linear combinations of H_1, H_2 that reproduce the diagonal matrices on the left. Each result is the Cartan element H_α dual to the corresponding root: under the Killing form, $H_\alpha \cdot H_\beta = \alpha \cdot \beta$ for any pair of roots.

7.4 Root addition

For roots α, β with $\alpha + \beta$ also a root, $[E_\alpha, E_\beta]$ is proportional to $E_{\alpha+\beta}$. We compute $[E_{12}, E_{23}]$ explicitly:

$$E_{12}E_{23} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = E_{13}, \quad (42)$$

$$E_{23}E_{12} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = 0, \quad (43)$$

$$[E_{12}, E_{23}] = E_{13} - 0 = E_{13}. \quad (44)$$

This realizes the root addition $\alpha_{12} + \alpha_{23} = (2, 0) + (-1, \sqrt{3}) = (1, \sqrt{3}) = \alpha_{13}$.

For combinations that are not roots, $[E_\alpha, E_\beta] = 0$. For example,

$$E_{12}E_{13} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = 0, \quad E_{13}E_{12} = 0, \quad (45)$$

giving $[E_{12}, E_{13}] = 0$, consistent with the fact that $\alpha_{12} + \alpha_{13} = (3, \sqrt{3})$ has length-squared $9 + 3 = 12 \neq 4$ and is not in the A_2 root system. The full pattern of vanishing and non-vanishing brackets is given in Appendix A and matches the A_2 adjacency rule exactly.

8 Fundamental representation

The eight generators act on $\mathcal{H}_C = \mathbb{C}^3$ as 3×3 matrices by construction. To verify that this is the fundamental representation $\mathbf{3}$ of $\mathfrak{su}(3)$, we identify the three weights (eigenvalues of H_1, H_2 on the basis vectors $|i\rangle$):

$$H_1|1\rangle = +1|1\rangle, \quad H_2|1\rangle = \frac{1}{\sqrt{3}}|1\rangle, \quad \mu_1 = (1, \frac{1}{\sqrt{3}}), \quad (46)$$

$$H_1|2\rangle = -1|2\rangle, \quad H_2|2\rangle = \frac{1}{\sqrt{3}}|2\rangle, \quad \mu_2 = (-1, \frac{1}{\sqrt{3}}), \quad (47)$$

$$H_1|3\rangle = 0, \quad H_2|3\rangle = -\frac{2}{\sqrt{3}}|3\rangle, \quad \mu_3 = (0, -\frac{2}{\sqrt{3}}). \quad (48)$$

The three weights μ_1, μ_2, μ_3 form an equilateral triangle of side $|\mu_i - \mu_j|^2 = 4$ centered at the origin (since $\mu_1 + \mu_2 + \mu_3 = 0$). This is the weight diagram of the fundamental representation $\mathbf{3}$ of $\mathfrak{su}(3)$ [3, Section 13]. The matter paper's identification of the three skew pairs with QCD color charges is therefore the statement that the trapped node, viewed as a localized state in $\mathcal{H}_C$, transforms in the fundamental representation of the geometrically constructed $\mathfrak{su}(3)$.

9 Scope

The defect geometry alone gives, without further input:

1. Three colors, from the three skew-edge pairs of K_4 (Section 3).
2. The Weyl group S_3 , from the quotient S_4/V_4 (Section 4).
3. The rank-2 maximal torus T^2 , from the three valence-bond U(1) phases modulo the diagonal U(1) (Section 5).

4. Two Cartan and six root generators, totaling $8 = \dim \mathfrak{su}(3)$ (Sections 5–6).
5. The A_2 root system at the correct lengths and angles (Section 7).
6. The Cartan-Weyl structure constants of $\mathfrak{su}(3)$ (Appendix A).
7. The fundamental representation on the three colors (Section 8).

The construction does not deliver:

1. The QCD coupling g_s and its running. These come from the bond energetics of the lattice, which the present construction does not address.
2. Asymptotic freedom and Λ_{QCD} , both dynamical scales.
3. Confinement as a Wilson-loop area law. The matter paper Section 6 derives a static topological version of confinement at the metric wall $L/\sqrt{3}$; the dynamical area law of continuum QCD is a separate matter.
4. Higher generations. The construction is a first-shell defect classification; second-shell extensions and heavier flavors require the broader spectrum analysis of [2].

10 Conclusions

The trapped tetrahedral defect of the SSM contains the data of $\mathfrak{su}(3)$. Combinatorics gives the colors, S_4/V_4 gives the Weyl group, and Bell-pair phases give the maximal torus. The eight resulting operators reproduce the Cartan-Weyl structure of $\mathfrak{su}(3)$, with the A_2 root system at its standard geometry. None of the inputs are adjusted to match the gauge group; they are read off from a single trapped node in the FCC vacuum.

What this does not deliver is the dynamical content of QCD — the coupling, the running, Λ_{QCD} . Whether the SSM bond dynamics realizes those quantities is the obvious next question, and one we do not answer here.

Acknowledgments

This work builds on the matter paper [1] of the SSM.

Data availability

The Python script that verifies the closure of all 28 commutators on $\mathfrak{su}(3)$ and reproduces the structure-constant table of Appendix A is publicly available at https://github.com/raghu91302/ssmtheory/blob/main/verify_su3.py. No other data were generated or analyzed in this study.

A Commutator table

All $\binom{8}{2} = 28$ commutators among the eight operators close on the 8-dimensional space spanned by $\{H_1, H_2, E_{12}, F_{12}, E_{13}, F_{13}, E_{23}, F_{23}\}$. Each entry below is computed by direct matrix multiplication of the form $[A, B] = AB - BA$; the result is then expressed as a linear combination of the basis operators using the explicit matrix forms in Eqs. (14)–(20). Three sample calculations were shown in Section 7; the remaining 25 follow the same pattern.

$[H_1, H_2]$	$= 0$	$[H_1, E_{12}]$	$= +2E_{12}$
$[H_1, F_{12}]$	$= -2F_{12}$	$[H_1, E_{13}]$	$= +1E_{13}$
$[H_1, F_{13}]$	$= -1F_{13}$	$[H_1, E_{23}]$	$= -1E_{23}$
$[H_1, F_{23}]$	$= +1F_{23}$	$[H_2, E_{12}]$	$= 0$
$[H_2, F_{12}]$	$= 0$	$[H_2, E_{13}]$	$= +\sqrt{3} E_{13}$
$[H_2, F_{13}]$	$= -\sqrt{3} F_{13}$	$[H_2, E_{23}]$	$= +\sqrt{3} E_{23}$
$[H_2, F_{23}]$	$= -\sqrt{3} F_{23}$	$[E_{12}, F_{12}]$	$= H_1$
$[E_{12}, E_{13}]$	$= 0$	$[E_{12}, F_{13}]$	$= -F_{23}$
$[E_{12}, E_{23}]$	$= E_{13}$	$[E_{12}, F_{23}]$	$= 0$
$[F_{12}, E_{13}]$	$= E_{23}$	$[F_{12}, F_{13}]$	$= 0$
$[F_{12}, E_{23}]$	$= 0$	$[F_{12}, F_{23}]$	$= -F_{13}$
$[E_{13}, F_{13}]$	$= \frac{1}{2}H_1 + \frac{\sqrt{3}}{2}H_2$	$[E_{13}, E_{23}]$	$= 0$
$[E_{13}, F_{23}]$	$= E_{12}$	$[F_{13}, E_{23}]$	$= -F_{12}$
$[F_{13}, F_{23}]$	$= 0$	$[E_{23}, F_{23}]$	$= -\frac{1}{2}H_1 + \frac{\sqrt{3}}{2}H_2$

The simple roots in the (H_1, H_2) basis are $\alpha_1 = \alpha_{12} = (2, 0)$ and $\alpha_2 = \alpha_{23} = (-1, \sqrt{3})$, with inner product $\alpha_1 \cdot \alpha_2 = -2$ and lengths $|\alpha_1|^2 = |\alpha_2|^2 = 4$. The Cartan matrix is computed as

$$A_{ij} = \frac{2\alpha_i \cdot \alpha_j}{\alpha_j \cdot \alpha_j}, \quad A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad (49)$$

the standard Cartan matrix of the A_2 Dynkin diagram, confirming the algebra as $\mathfrak{su}(3)$.

References

- [1] R. Kulkarni, *Matter as Incomplete Crystallization: Quark Charges, Color Confinement, and the Proton Mass from a Single Extra Node in the Vacuum Lattice*, SSMTheory Group, IDrive Inc., Calabasas, CA (2026). Manuscript number PHYSO-D-26-00185 (in revision at Phys. Open); preprint: [doi:10.5281/zenodo.18917946](https://doi.org/10.5281/zenodo.18917946).
- [2] R. Kulkarni, *The mass-energy-information equivalence: A bottom-up identification of the particle spectrum via FCC lattice error correction*, Phys. Open (in press, 2026). [doi:10.1016/j.physo.2026.100414](https://doi.org/10.1016/j.physo.2026.100414).
- [3] J. E. Humphreys, *Introduction to Lie Algebras and Representation Theory*, Springer (1972).
- [4] H. Georgi, *Lie Algebras in Particle Physics*, 2nd ed., Westview (1999).
- [5] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, 10th anniversary ed., Cambridge University Press (2010).